

Unit 3, Lesson 8: Solving Real-World Problems Involving Numbers Expressed in Scientific Notation

Benchmark

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Learning Targets

- ☐ I can solve real-world problems involving operations with numbers expressed in scientific notation.
- ☐ I can express answers in scientific notation using significant digits.

1. Using the digits 1 – 9 at most twice each, create the sum of the four expressions that has the greatest possible value.

$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	+	$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	=
$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	-	$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	=
$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	x	$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	=
$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	÷	$\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix} \times 10^{\begin{bmatrix} \square \\ \square \end{bmatrix}}$	=
Total			

3.8.2 Guided Instruction: Understanding Significant Digits in Scientific Notation

1. The table shows the distance from Earth, in kilometers (km), to the nearest two planets.

Planet	Distance (km)	Scientific Notation
Venus	38 million	3.8×10^7
Mars	55.7 million	5.57×10^7

Jackson was asked to find the difference between the two distances by rounding to the nearest whole number to make the values easier to work with.

$$(6 \times 10^7) - (4 \times 10^7) = (2 \times 10^7)\text{km}$$

Jocelyn was asked to perform the calculation without rounding.

$$(5.57 \times 10^7) - (3.8 \times 10^7) = (1.77 \times 10^7)\text{km}$$

- a. What is the difference, in km, between the two answers?
- b. Discuss with your partner why Jocelyn’s answer is more precise.



Significant digits are non-zero digits of a number and the zeros that are included between them or any trailing zeros that are considered to be precise.

- c. Based on the definition, how many significant digits are in the representations of the distances for Mars and Venus?

Examples	
2027 has 4 significant digits.	00052000 has 2 significant figures.
00.00000 has 3 significant digits.	00035000.0 has 6 significant figures.

There are three rules for determining how many significant figures are in a number:

- Non-zero digits are always significant.
- Any zeros between two significant digits are significant.
- A final zero or trailing zeros in the decimal portion **ONLY** are significant.

Sometimes answers written in scientific notation are written using a specific number of significant digits. If two numbers are written with different precision, then the precision is the least number of significant digits.

- d. Discuss with your partner the given values of the distance from Earth to Mars and Venus. Determine how many significant digits should be used to represent the difference between Mars and Venus.

2. Circle the significant digits in each number. Place a check in the column(s) to reflect which rule(s) you used when making your decision.

	Non-Zero Digits Always Significant	Zeroes Between Non-Zero Digits Always Significant	Zeroes to the Left of Non-Zero Digits Not Significant	All Zeroes to the Right are Significant if There is a Decimal Point
0.0807				
13,401				
1,990				
5.5				
367.0				

3. In 1966, the Apollo Guidance Computer was developed to make the calculations that would one day put humans on the Moon. In 2020, one phone company released the 12th smartphone in its latest series of smartphones.

Information about the memory chips in the devices is summarized in the table.

Device	Storage	Processing	Memory
Apollo 11 Guidance Computer	32 KB	1.042 MHz	2 KB
Smartphone	128 GB	2.7 MHz	6 GB
*For reference, storage and memory are measured in bytes (B), and processor speed is measured in hertz (Hz)			

Kilobytes (KB) = 1,000 bytes (B)

Megahertz (MHz) = 1,000,000 hertz (Hz)

Gigabytes (G) = 1,000,000,000 bytes (B)

Terabytes (T) = 1,000,000,000,000 bytes (B)

- a. Determine the combined storage and memory, in bytes, of the smartphone. Write your answer using scientific notation.**

- b. How much combined storage and memory, in bytes, are in 8 smartphones? Write your answer using scientific notation.**

- c. Determine how much faster, in hertz, the smartphone's processing speed is compared to the Apollo 11 Guidance Computer's processing speed. Write your answer using scientific notation.**

3.8.3 Guided Instruction: Using Scientific Notation to Solve Real-World Problems

1. The National Debt in 2020 was \$26.9 trillion. The population for the state of Florida in 2020 was estimated to be 21.9 million.
 - a. If the people living in Florida were to pay off the national debt, how much would each person pay?
 - b. The population of North Bay in 2020 was 197,974. If the people living in North Bay were to pay off the national debt, how much would each person pay?
 - c. It has been estimated that the national debt increases at a rate of about \$3.9 million each day. At this rate, how much will the debt increase in a year?

3.8.4 Your Turn!

1. Jennifer Lopez changed the Hollywood landscape, single-handedly increasing Latino representation in film, television, and fashion. J.Lo, as she is known, has become one of the highest-paid Latina celebrities in Hollywood. She has multiple income streams.

The table summarizes her income from some of her recent jobs.



Job	Income
Judge and executive producer on the show "World of Dance" for 4 seasons	1.2×10^7 dollars per season
Jennifer Lopez clothing line, cosmetics, and fragrances	3×10^8 dollars in revenue
2012 "Dance Again" World Tour	5.1×10^7 dollars
2016 – 2018 "All I Have" concert residency in Las Vegas, Nevada	3.5×10^5 dollars per season

- a. What is the difference in Jennifer Lopez's income from the "Dance Again" World Tour and 2016 – 2018 concert residency in Las Vegas, Nevada?
- b. What is the sum of Jennifer Lopez's income from "World of Dance" and her clothing line, cosmetics, and fragrances?

- c. How many times greater is Jennifer Lopez's income from her clothing line, cosmetics, and fragrances compared to the income she made on her 2012 World Tour?
- d. Determine if Jennifer Lopez's income from 2016 – 2018 concert residency in Las Vegas, 2012 "Dance Again" World Tour, and show "World of Dance" is larger than her income from her clothing line, cosmetics, and fragrances.
2. There are about (2.0×10^{11}) visible stars in a galaxy. There are about (8.0×10^{10}) galaxies in the universe. How many visible stars are there in the universe? Write your answer in scientific notation.

3.8.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Writing large numbers in scientific notation			
Writing small numbers in scientific notation			
Adding and subtracting numbers expressed in scientific notation			
Multiplying numbers expressed in scientific notation			
Dividing numbers expressed in scientific notation			
Solving real-world problems involving operations with numbers expressed in scientific notation			